Choosing How to Respond to Innovation Paradoxes

From Strategic Choices to Organisational Value Creation

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When Understanding Becomes a Leadership Challenge

The management team has made progress. It understands why two competing convictions about innovation both deserve serious consideration. It has explored four possible ways of responding to their tension. Yet the most important question remains unanswered: What should the organisation actually do?

In Part 1, we examined Capability Push versus Market Pull: whether distinctive organisational capabilities or customer needs should take precedence in innovation. Part 2 introduced four response directions: Integration, Contextual Differentiation, Dynamic Alternation and Explore a Synthesis. Understanding these options does not establish which deserves priority.

Leadership must consider what the organisation wants to achieve, the competitive challenges it faces, its capabilities and the uncertainty it can accept. The Innovation Paradox Thinking Toolkit supports this transition through Position, Prioritise, Design and Learn & Adapt. (Teece, 2007; Klokgieters, 2020).

A Company at a Crossroads

Consider a fictional industrial technology company employing approximately 500 people. It develops and manufactures sophisticated machines. Its engineering expertise and customer relationships underpin its success. Innovation has typically started either with new technological capabilities developed by R&D or with customer requirements translated into product improvements.

Customers now demand higher machine availability, lower maintenance costs and fewer unexpected production interruptions. Meanwhile, sensor technology, software and data analytics offer ways to anticipate equipment failures and prevent downtime. (Baines et al., 2009).

The Commercial Director argues: “Our customers want reliable machines and faster service. We know where the problems are. Why divert scarce resources towards technologies whose commercial value is uncertain?”

The Head of R&D replies: “Our emerging capabilities could fundamentally change how we support customers. If we focus exclusively on today's requirements, we may miss tomorrow's opportunities.”

Both arguments are legitimate. The team has analysed the paradox and explored its response options. Now it must decide. This storyline is a constructed simulation, not an observed case study.

1. Position: Understanding the Strategic Possibilities

The management team compares each response along two dimensions. Value Creation Potential considers strategic impact, innovation performance and long-term adaptability. Ease of Implementation considers capability readiness, organisational acceptance and implementation effort. The positioning matrix makes

assumptions visible; it does not automatically rank the alternatives. Any placement of the responses is illustrative and specific to this fictional company, not a universal assessment.

Integration looks accessible: a multidisciplinary project could bring engineering, software, service and commercial expertise together to develop predictive maintenance functionality using existing structures.

Contextual Differentiation would give Market Pull greater influence over improvements to established machines, while allowing Capability Push to lead selected exploratory activities. It is feasible, although classifying opportunities and allocating resources could prove difficult.

Dynamic Alternation would allow the leading perspective to shift deliberately when new evidence or circumstances justify it. The team doubts whether current decision-making processes can yet support such shifts consistently.

Explore a Synthesis would investigate a semi-permanent organisational capability dedicated to Maximising Customer Equipment Uptime. Engineering, data analytics, service and customer knowledge would be organised around continuing value rather than a single project deliverable. Its strategic potential is intriguing, but its governance and resource implications are uncertain.

The options occupy different positions in the matrix. Nevertheless, an attractive matrix position is not the same as a strategic priority.

2. Prioritise: Strategy Changes the Decision

“Integration offers clear benefits and is relatively easy to implement,” says the Commercial Director. “Why don't we concentrate our resources there?”

“Because improving today's products may not be enough,” responds the Head of R&D. “Competitors are building connected equipment and data-driven services.”

The CEO reframes the discussion: “Are we trying to build better machines, or should we organise ourselves to maximise the value customers obtain from using them?”

Customers invest in machines to secure reliable production capacity. Unplanned downtime reduces that capacity and creates costs. Equipment availability is therefore a tangible source of customer value. The question is whether the company's existing innovation arrangements are sufficient to deliver it continuously.

The team considers strategic ambition, competitive dynamics, consequences of inaction, timing and the value of preserving future options. It must protect current customer relationships without ignoring emerging service opportunities. It can implement some improvements quickly while using a bounded experiment to investigate a different organising principle. (March, 1991).

The resulting portfolio of choices is deliberate. Integration is prioritised for improving innovation in the existing business. Contextual Differentiation is introduced in selected exploratory areas. A protected, limited investment is allocated to Explore a Synthesis around customer equipment uptime. Organisation-wide Dynamic Alternation is deferred until management processes become more capable of supporting it.

This is not an equal split between competing interests. It is a set of priorities reflecting different objectives, time horizons and risks. The positioning of a response informs leadership judgement; it does not replace it.

3. Design: Turning Choices into Organisational Action

For Integration, the company establishes a project to develop predictive maintenance functionality. Engineering provides sensing and diagnostic expertise; software specialists build analytical models; service contributes

failure and maintenance knowledge; and commercial specialists test customer relevance. The project has a defined scope, budget and responsibilities. Resources are allocated from existing departments and released or reassigned when the solution is transferred into regular operations.

For Contextual Differentiation, the company defines criteria distinguishing established-product improvements from exploratory technology development. These criteria clarify when customer evidence should lead and when emerging capabilities merit greater freedom.

Explore a Synthesis requires a different design hypothesis. The company proposes a small, semi-permanent capability with the mission: Continuously maximise customer equipment uptime.

This capability combines engineering, software, data analytics, service and customer knowledge. It is not responsible for one predefined solution. It continually identifies, develops and implements opportunities to improve machine availability, whether through remote diagnostics, predictive maintenance, service redesign or new equipment-support offerings.

The capability needs decision authority across functional boundaries, continuity of resources and performance measures linked to sustained outcomes. These might include machine availability, unplanned downtime, customer satisfaction and economically sustainable value for the customer and the company.

Its continuation depends on demonstrated value, not merely the passage of time. If the contribution declines, leadership must reconfigure, redirect or discontinue it. A permanent team is not automatically a synthesis: the arrangement must genuinely shift the organising principle from temporary functional resource allocation towards an enduring customer-value mission.

The team defines a testable hypothesis: A semi-permanent capability dedicated to maximising customer equipment uptime can create more sustained customer value than repeatedly assembling temporary innovation projects within the existing functional structure.

The experiment receives an accountable leader, limited resources, decision boundaries and an initial review date. If it only improves cross-functional collaboration, it may represent valuable Integration rather than a genuine synthesis. A new organisational label is not evidence of a new organisational capability.

4. Learn & Adapt: Strategic Choices Are Not Final Answers

Six months later, the management team holds its first structured review. This is an early checkpoint for testing feasibility, organisational learning and initial customer signals, not a deadline for proving durable value creation. Demonstrating whether the capability creates sustained benefits will require evidence over a longer period.

Performance Learning asks whether innovation outcomes and customer equipment availability are improving. Design Learning examines whether responsibilities, resources and organisational arrangements work as intended. Strategic Learning tests whether the original assumptions and response priorities remain appropriate.

Imagine mixed early signals. The predictive maintenance project is progressing, but resource tensions persist. Contextual Differentiation clarifies several decisions, although some opportunities resist simple classification. The uptime capability has identified promising customer opportunities, but funding and authority remain contested. These are illustrative possibilities, not measured results.

The team must distinguish promising activity from realised value. It asks whether machine availability is improving, whether the capability adapts more effectively than conventional project arrangements and whether any benefits justify its organisational complexity. Six months may provide evidence about feasibility and learning, but not yet establish durable performance superiority.

The team may continue, modify, expand or stop the experiment. It may even revisit its understanding of the original paradox. Treating a response as a hypothesis that must earn its place is different from declaring it a permanent solution.

What the Management Team Has Achieved

The team began with a disagreement about whether capabilities or markets should drive innovation. Rather than settle for an unexamined compromise, it analysed the convictions, compared four response directions, made strategic priorities explicit and translated selected choices into organisational experiments.

The question of equipment uptime made the potential synthesis concrete. It did not prove that a semi-permanent capability would outperform project-based Integration. It established why the possibility was important enough to investigate, what would need to change and what evidence could challenge the hypothesis.

Leadership does not need certainty about the best solution before deciding what deserves exploration. It needs sufficient understanding to make deliberate choices, design credible tests and learn from their consequences.

Leadership Beyond the Paradox

Innovation paradoxes will continue to emerge as technologies, markets and organisational capabilities change. Leadership cannot eliminate every contradiction, but it can develop a stronger capability for working with competing convictions.

Across this series, Part 1, Understanding Innovation Paradoxes, examined how to analyse a tension. Part 2, Responding to Innovation Paradoxes, introduced four response directions. Part 3, Choosing How to Respond to Innovation Paradoxes, has shown how leaders can evaluate options, set priorities, design organisational arrangements and adapt through learning.

The Innovation Paradox Thinking Toolkit does not provide a formula for resolving every paradox. It offers a structure for making assumptions explicit, expanding strategic options and acting deliberately.

Innovation paradoxes become strategically valuable when leadership turns competing convictions into better choices, purposeful organisational action and continuous learning.

AUTHOR'S NOTE | The management-team storyline is a fictional analytical simulation. Its choices and early observations are hypothetical and do not constitute empirical evidence of organisational performance.

ABOUT THIS SERIES | Innovation Paradox Thinking is a three-part series exploring how leaders can understand competing convictions, examine possible responses and make strategic choices that support organisational value creation.

Visual companion

From Strategic Choices to Value Creation | The four leadership activities in one view

Illustrative framework: the positioning of responses and six-month observations relate to a fictional company. They are not universal rankings or empirical findings. Six months represents an initial learning checkpoint, not proof of sustained value.

About the Author

Koen Klokgieters is the founder of InnoPlot and has spent more than three decades working at the intersection of strategy, innovation, technology and organisational transformation. His experience combines corporate leadership, consulting, entrepreneurship and education.

Through InnoPlot, he supports organisations and their leaders in translating strategic challenges into innovation capabilities, organisational change and measurable impact.

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Main visual | Innovation Paradox Thinking



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